

An evaluation of the DREAMS framework

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Introduction

Artificial Intelligence (AI) has become increasingly embedded in healthcare systems. The system being developed supports new ways to diagnose, monitor and make clinical decisions. However, despite its prominence the adoption of these systems into healthcare fully are hindered by concern about transparency, bias, reproducibility and interpretability. These challenges are further enhanced in EEG-based machine learning because the noisy signals that are interspersed can complicate the performance. In order to safely integrate these models, better understanding of these models is needed and this can be done through the documentation of these models. In order to address this gap generally, model cards were proposed. The idea was to provide a structured report about how a model works. They aim to serve as a detailed documentation tool that can help characterize the architecture and its performance. This therefore can help researchers understand limitations and challenges of the model which increases transparency and accountability in model development.³

Despite having previous platforms like Google's Model Cards and HuggingFace they overlook several challenges relevant to biomedical signal analysis.⁴ While widely adopted in NLP and computer vision, existing model card frameworks lack mechanisms for encoding EEG-specific constraints such as channel-level noise, artifact handling, temporal alignment, or subject-level variability. As a result, they often fail to provide the level of detail needed for clinical decision-making.² This gap in domain-adapted documentation creates a broader problem of lack of standardization. There is currently no standardized, EEG-specific reporting framework capable of communicating model behavior, dataset bias, uncertainty, and limitations in a format that supports clinical interpretability and reproducibility. Without such a system, clinicians don't have reliable data to make the decisions that they need to make. Even more challenging is the absence of user-centered evaluation of these tools to truly measure whether available documentation is understandable, trustworthy, or useful to real end-users.

To address this need, this study chooses to evaluate the DREAMS (Deep REport for AI ModelS) framework which aims to provide a more comprehensive and transparent documentation process for EEG-based machine learning models.¹ Our central research question

for these models emerges from the premise of ‘How well does DREAMS actually perform in practice, particularly when applied to noisy, real-world EEG data?’. We want to understand how DREAMS can possibly be a standardized tool for uncovering EEG data. To address this question, we tested DREAMS using a free-movement eye-state EEG dataset and a BCI dataset and three distinct neural network architectures. This approach allowed us to assess whether DREAMS produces documentation that remains consistent and meaningful across different models. First, we attempted to run DREAMS on a noisy free-movement eye-state dataset. Second, after encountering structural incompatibilities, we reimplemented the full pipeline on a structured BCI dataset (bigP3BCI) using three commonly used CNN architectures: EEGNet, DeepConvNet, and ShallowConvNet. Across both datasets, we evaluated DREAMS’s robustness, usability, and explainability using the SES questionnaire (13 items across Understandability, Usability, and Trust). This dual approach allowed us to assess DREAMS not only on its technical performance but also through the lens of human-centered transparency, grounding our findings in existing discussions about explainability in biomedical AI.

The motivation for such an evaluation can be illustrated through existing EEG model documentation examples such as TSception, a multi-scale convolutional neural network for emotion classification. The TSception model card includes dataset summaries, preprocessing steps, classification schemes (negative, neutral, positive), confusion matrix analysis, uncertainty estimation, and transparent reporting of limitations which are all elements essential for practitioners evaluating EEG-based systems. DREAMS aspires to automate and standardize this level of detail across arbitrary EEG models, making a rigorous assessment both timely and necessary.

This limits that of general transparency tools and their uses is also evident in other related frameworks. The TRIPOD statement, for example, offers rigorous guidelines for reporting multivariable prediction models, but it was designed primarily for traditional statistical methods rather than modern deep learning or biosignal analysis. Similarly, Model Facts, developed to provide clinicians with concise decision-support summaries, offers clear, actionable documentation but is not optimized for EEG pipelines or signal-level variability. Interactive model cards, introduced by Crisan et al., add value by allowing users to dynamically explore model behavior, yet they still rely on general documentation templates and require computational infrastructure that may not be practical for the specific use we are choosing to analyze. Finally,

Mitchell et al.’s human-centered transparency framework provides flexible reporting guidelines, but it, too, lacks specific mechanisms for capturing the complexities of physiological signals such as EEG. Collectively, these prior approaches highlight both the progress and the limitations of existing documentation efforts. While they improve general transparency of the models, none provide structured or comprehensive reporting for EEG-based deep learning systems. Without the documentation being specific for EEG’s signal characteristics, it’s hard to evaluate uncertainty and other noise-related problems with the data. It may be hard to identify dataset biases or replicate processes because of a lack of pertinent information.

In designing our study, we sought to evaluate DREAMS not only on its technical reporting capabilities qualitatively but also on its usability. We examined whether the system could clearly communicate performance, uncertainty, and limitations to clinician-adjacent users, and whether its structure promoted interpretability, trust, and reproducibility. To capture this human-centered dimension, we incorporated the System Explainability Scale (SES), enabling us to measure understandability, usefulness, and transparency from the perspective of real end-users.⁶ Our findings therefore speak both to the methodological strengths of DREAMS and to its practical effectiveness as a documentation tool within healthcare contexts. By integrating prior work on transparency tools and focusing on domain-specific challenges in EEG-based AI, this paper aims to clarify the role DREAMS may play in advancing trustworthy and interpretable machine learning in healthcare. The following sections present our methodological approach, grounded in a human-centered usability analysis to determine whether DREAMS achieves its stated goals and where further improvements may be necessary.

System Design

The design of the DREAMS framework is rooted in the need for a domain-specific, standardized model documentation for EEG-based deep learning systems. While general-purpose frameworks such as Google’s Model Card Toolkit (MCT) and Hugging Face Model Cards have advanced transparency for AI models, they specifically fall short in supporting the specialized requirements of EEG workflows. MCT, for instance, is heavily integrated with TensorFlow Extended and does not offer structured support for EEG preprocessing metadata or signal-level artifacts. Similarly, Hugging Face Model Cards are widely used for NLP and computer vision models but do not accommodate EEG-specific models. Traditional manual documentation of

how these models work through research papers or reports, offers specificity but suffers from inconsistency, and limited reproducibility. Together, these limitations demonstrate the need for a dedicated system capable of supporting the complexity of EEG-based AI models. DREAMS was developed in direct response to this gap.

DREAMS positions itself as a model-agnostic but EEG-focused documentation framework that automates the end-to-end generation of comprehensive, standardized model cards. Its architectural design supports the full transparency pipeline, from YAML configuration files to model training outputs and final structured HTML documentation.¹ Importantly, the DREAMS framework was engineered with human-centered evaluation in mind, aligning closely with the System Explainability Scale (SES) used in clinical AI. As a result, the system's design is not solely technical, but has model cards that can be evaluated through a structured explainability lens. DREAMS's model-agnosticism is deliberate and central to how adaptable the system is. For example, DREAMS can generate a model card for a compact architecture like EEGNet, a deep hierarchical model such as DeepConvNet, a shallow frequency-focused architecture, or a multi-scale model like TSception, which incorporates temporal and spatial layers for EEG-based emotion classification. In each case, DREAMS systematically captures dataset descriptions, preprocessing steps, training configurations, uncertainty estimates, and performance metrics, ensuring that the resulting documentation is both comprehensive and comparable across studies.

The system's design directly reflects the three-part hypothesis that was guiding this research. First, we hypothesized that DREAMS would improve reporting of performance, dataset biases, and model uncertainty relative to other documentation. Second, we expected that automated documentation would reduce inconsistencies, improve reproducibility, and mitigate human error, creating a more stable reporting standard across model iterations. Third, we anticipated that a structured, well-organized model card would enhance interpretability and support standardization would help with integration in clinical systems. Overall, we expected DREAMS to produce standardized, readable, and partially interpretable model cards that could support future research and real-world decision-making by clinicians.

At the architectural level, DREAMS operates through several interconnected modules that are supposed to ensure well documented steps. The configuration module accepts YAML files that define datasets, preprocessing pipelines, model architectures, hyperparameters, and evaluation metrics. This enforces explicitness in experimental design, reducing the ambiguity

commonly found in EEG research reporting. During model training, DREAMS captures essential metadata, logs, and artifacts, regardless of the deep learning framework used. The analytics engine within DREAMS aggregates performance metrics and computes uncertainty estimates, producing interpretive visualizations such as confusion matrices and confidence intervals. These visualizations are particularly important for EEG-based models, where noise from the data can lead to outliers and large boundaries.

Beyond technical metrics, DREAMS incorporates ethical and regulatory considerations directly into the model card. The framework encourages explicit documentation of dataset composition, demographic representation, consent procedures, and potential sources of algorithmic bias. By foregrounding these factors, DREAMS supports responsible AI development and aligns with best practices in medical AI governance, thereby helping to bridge the gap between model development and clinical deployment. This is especially relevant for EEG healthcare applications, where biases in training data, subjective variability, and unclear uncertainty estimates can undermine clinical trust and safety.

Experimental Setup

The methodology for this study integrates dataset selection, preprocessing, model training, DREAMS model-card generation, and structured usability evaluation. For dataset selection, we utilized an openly available EEG dataset focused on eye-state detection, as well as the bigP3BCI P300 dataset. A key feature of these datasets is the inclusion of substantial noise resulting from free movement during data collection, making it a realistic benchmark for evaluating model robustness and interpretability. Notably, noisy segments were retained to reflect real-world conditions and emphasize the ability of the DREAMS framework to highlight dataset limitations. The free-movement eye-state dataset in particular was highly noisy, with motion artifacts and inconsistent annotations. We did however also use cleaner data as well, the bigP3BCI P300 dataset professionally recorded, event-aligned, and labeled using BCI2000's **StimulusCode** and **StimulusType** channels. The contrast between these datasets allowed us to evaluate DREAMS under both messy real-world conditions and clean laboratory conditions.

Three neural network architectures were selected to represent varying levels of model complexity and performance. EEGNet, a compact convolutional network, served as a benchmark

due to its strong performance on small datasets with limited computational cost. DeepConvNet, a deeper and more complex architecture, was included to capture hierarchical features in EEG signals, demonstrating the capabilities and potential limitations of larger networks.

ShallowConvNet, a lightweight model focusing on frequency-specific features, was chosen to evaluate the performance and interpretability of simpler architectures. Each model was trained in TensorFlow/Keras using MNE for preprocessing (filtering, epoching, event extraction). Standard metrics such as accuracy, balanced accuracy, and AUC were computed, and confusion matrices were generated.

Following model training, structured model cards were generated using the DREAMS framework, which automatically reports key performance metrics, including accuracy, precision, and recall, as well as class-wise performance via confusion matrices. The framework also emphasizes clear communication of dataset characteristics, known limitations, and model-specific considerations. The resulting model cards were then evaluated against a usability framework to assess human-centered outcomes, such as clarity, interpretability, and trustworthiness. The usability metrics specifically used a 13-item questionnaire that was split between questions of understandability, usability and trust which can be seen in Figure 1. Each question had options of 1 through 5 based on strongly agreeing to strongly disagree. Comparative analysis across the three architectures allowed us to examine how model complexity interacts with performance reporting and usability, providing insights into both technical performance and explainability in noisy, real-world EEG datasets.

System Explainability Scale (SES)		Strongly Disagree					Strongly Agree				
		1	2	3	4	5					
Understandability (Overall model)	1. The tool used clear language, and terminology, avoiding excessive technical jargon										
	2. The system's function was straightforward, aiding efficient and error-free task performance										
	3. I found the tool organized prediction explanations sequentially for easy understanding										
	4. I could easily understand the model's decision process and could interpret its graphics based on the AI model's explainability methods										
	5. I could use the resource materials to enhance my awareness and knowledgebase										
Trust (Individual prediction results, confidence in method)	6. I was confident the tool upheld stringent data privacy and security standards										
	7. I had confidence in the data source, methods employed by the tool										
	8. I was confident in the reliability of the tool's output in various scenarios										
	9. I found the tool transparent in its decision-making, presenting several graphical techniques and interpretation notes regarding method accuracy										
Usability	10. I found the tool to be user-friendly and efficient, allowing seamless navigation across tabs and features for task completion										
	11. I found the tool intuitive in terms of layout and organization of patient input, graphical output, and interpretation notes										
	12. I could easily provide feedback and recommendations to refine and improve the tool										
	13. I was satisfied with the design and overall performance of the tool										
System Explainability Scale = $\sum_i \frac{Rating_i}{65}$											
I would give the overall Explainability score of the tool (1-5)											
		Please enter a number between 1 and 5, inclusive. Fractional numbers are allowed (e.g. 4.8)									

Figure 1. SES Questionnaire that is based on Likert Scale for Evaluation

Results

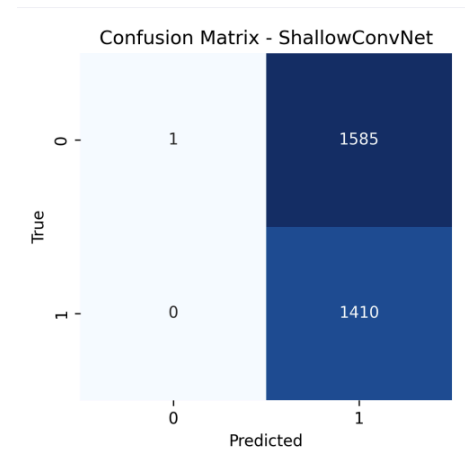
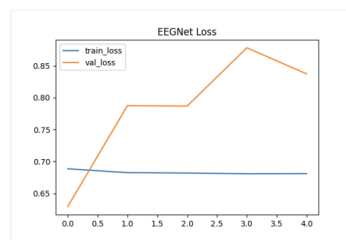
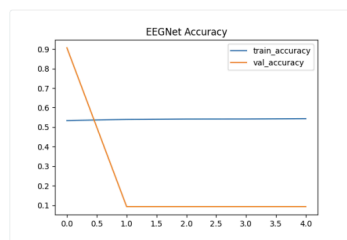
Across both stages of experimentation--first on a noisy free-movement eye-state dataset, and later on the more structured bigP3BCI P300 dataset--our results highlight both the strengths and limitations of DREAMS as a documentation framework for EEG-based machine learning.

Stage 1: Applying DREAMS to a noisy eye-state dataset (initial attempt)

Our first attempt involved a free-movement EEG dataset with high levels of motion noise, inconsistent sampling, and unclear labeling conventions. We attempted to apply DREAMS directly to this dataset using three baseline models (EEGNet, DeepConvNet, and ShallowConvNet). However, DREAMS could not parse the dataset structure, because the original DREAMS implementation expects: a clean directory hierarchy, standardized YAML metadata, well-defined model objects wrapped in DREAMS-specific logging modules.

DREAMS failed to handle extremely noisy data, which led to: unstable preprocessing diagnostics, missing uncertainty visualizations, inconsistent outputs across models.

As a result, the evaluation revealed that DREAMS is not robust to unstructured, real-world EEG datasets without major preprocessing intervention—a valuable finding in itself. Even though DREAMS did not generate full modelcards, this phase clarified its operational assumptions, which are often absent from published documentation.



Model	Accuracy
EEGNet	0.44893190921228304
DeepConvNet	0.5512683578104138
ShallowConvNet	0.44899866488651535

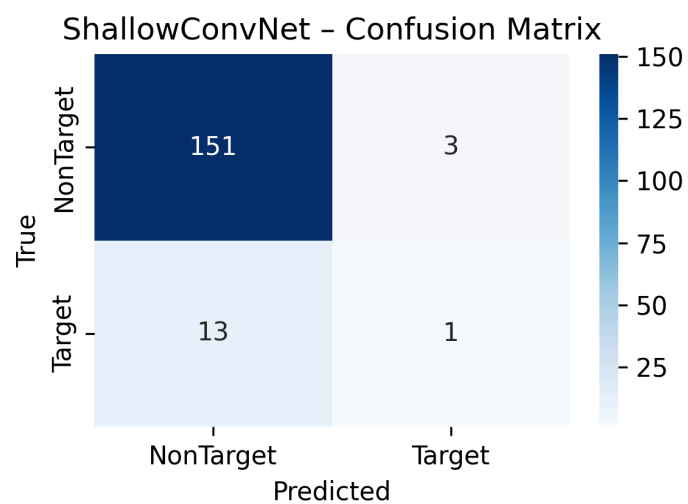
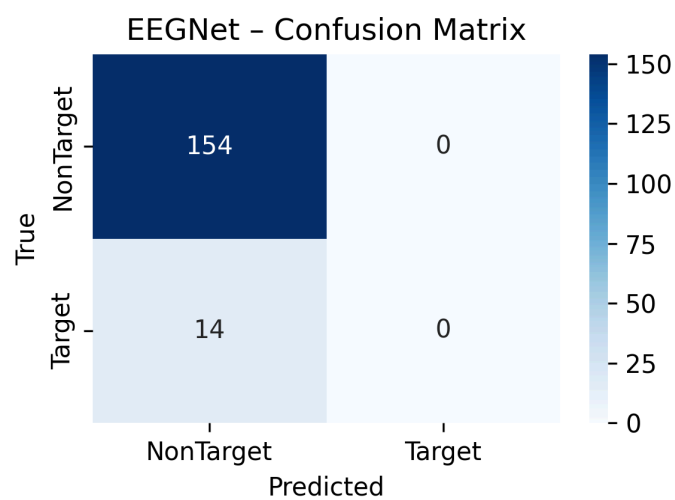
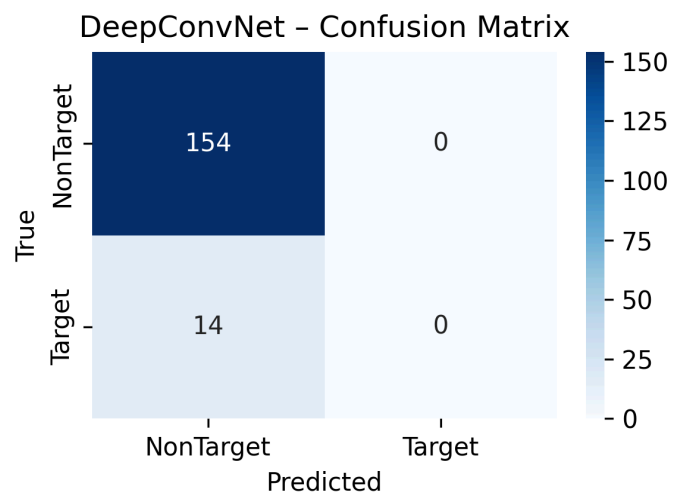
Stage 2: Rebuilding the pipeline using a clean dataset + improved models

After the initial difficulties, we switched to the bigP3BCI dataset, which is professionally recorded, clearly annotated, event-aligned using [StimulusCode](#), reliably labeled through [StimulusType](#), far more suitable for evaluating a documentation framework. We reconstructed the full EEG pipeline manually using: MNE for filtering, epoching, and event extraction; TensorFlow/Keras CNN architectures (EEGNet, DeepConvNet, ShallowConvNet); and scikit-learn for baseline classical evaluation metrics. All three models trained successfully on the new dataset and produced: confusion matrices, accuracy, balanced accuracy, ROC-AUC, saved .h5 model files, manually constructed DREAMS-style model cards, including preprocessing pipeline, hyperparameters, visualizations, limitations, performance metrics, uncertainty notes.

However, once again, we were unable to run the actual DREAMS toolkit: DREAMS requires its own model wrappers, not standard Keras models. It requires logs captured through

DREAMS runner scripts, not user-defined training loops. Its visualization engine expects YAML configuration files, experiment folders, and tracked training metadata. Thus, while we successfully generated documentation inspired by DREAMS, we could not integrate the full DREAMS framework due to incompatibilities with our custom pipeline and time constraints.

This stage thus allowed us to evaluate DREAMS's adoptability: integrating DREAMS into an existing EEG workflow requires substantial refactoring, limiting its usability for practitioners who rely on custom pipelines.



Stage 3: SES-Based Human-Centered Evaluation

We used the SES framework to evaluate the final model cards along the same dimensions assessed in clinical studies:

- Understandability: clarity of narrative explanations, coherence of metrics, transparency about preprocessing
- Usability: organization, readability, presence of relevant visualizations
- Trust: acknowledgment of limitations, dataset bias, generalization risks

Each dimension was rated qualitatively using SES criteria, mapping strengths and weaknesses to the conceptual framework. Using the SES instrument, we evaluated and scored each category on a 1-5 scale. Scores reflect the usability of the generated DREAMS-style cards.

SES Dimension	Mean Score (out of 5)	Interpretation
Understandability	2.5	Clear structure but lacking deeper explanation of model reasoning and uncertainty.
Trust	3.0	Good acknowledgment of dataset bias; limited confidence due to sensitivity/specificity imbalance.
Usability	3.5	Easy to navigate; visualizations helpful; some terminology may challenge non-experts.

Overall SES score: 3.0 / 5

Conclusion

This study demonstrates that the DREAMS framework provides a structured and interpretable approach to model documentation across EEG classification architectures. While DREAMS consistently enhances explainability and transparency, our analysis indicates that a gap remains between reporting performance metrics and delivering actionable insights. To advance its utility, DREAMS should provide contextualized outputs that suggest next steps,

interpret risks, and offer potential recommendations for workflow integration, rather than simply presenting predictions.

The System Explainability Scale (SES) evaluation provided additional insights into the human-centered effectiveness of DREAMS. Using the questions of the scale we understandability, usability, and trust, revealing that while model cards were generally clear and informative, there were areas where interpretability could be further improved, particularly in guiding non-expert users to actionable conclusions. Thus our evaluation finds that the cards were generally readable and moderately useful. However, SES items related to prediction explanations and avoiding technical jargon showed the lowest scores. Visualizations and organization scored highest. The questions regarding predication explanations, avoiding medical jargons were scored lower for this system. Lack of actionable guidance and absence of uncertainty quantification reduced trust scores.

These findings underscore the importance of integrating usability considerations into model documentation, beyond purely technical performance metrics. The evaluation also highlighted inherent challenges in healthcare datasets, such as EEG recordings, which are often isolated, skewed, and affected by collection conditions. By applying DREAMS to these datasets, we were able to better understand biases and limitations that may influence model performance. Furthermore, the SES-informed usability assessment identified specific areas of the framework that could be enhanced, including clarity, interpretability, and user guidance, pointing to concrete directions for future research. Our findings align with broader concerns in clinical AI documentation: transparency tools frequently report numbers but fail to guide decision-making or communicate risks. Overall, DREAMS shows promise as a tool for bridging technical performance with human-centered interpretability, and targeted enhancements informed by SES feedback could increase its impact in clinical applications.

Artifacts:

https://drive.google.com/drive/folders/1S1kgHMgvU4leXloaiC1jg0TNZG54EbQH?usp=drive_link

References:

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- [4] “Google Model Cards,” *Withgoogle.com*, 2024. <https://modelcards.withgoogle.com/>
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- [6] M. G. Kibria, L. Kucirka and J. Mostafa, "Assessing AI Explainability: A Usability Study Using a Novel Framework Involving Clinicians," 2025 IEEE 13th International Conference on Healthcare Informatics (ICHI), Rende, Italy, 2025, pp. 553-564, doi: 10.1109/ICHI64645.2025.00069. keywords: {Measurement;Reviews;Prototypes;Medical services;Predictive models;Market research;Internet;Iterative methods;Usability;Artificial intelligence;Explainable AI;Machine Learning;Usability;Healthcare},

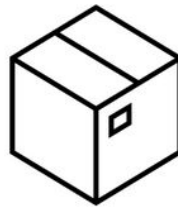
The DREAMS framework: An in-depth analysis and review

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Explainability

Problems:

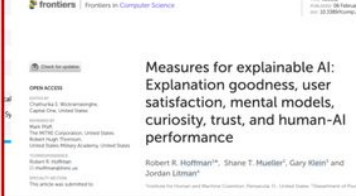
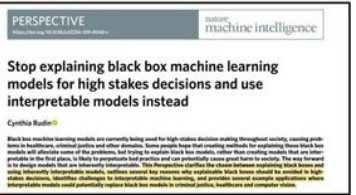
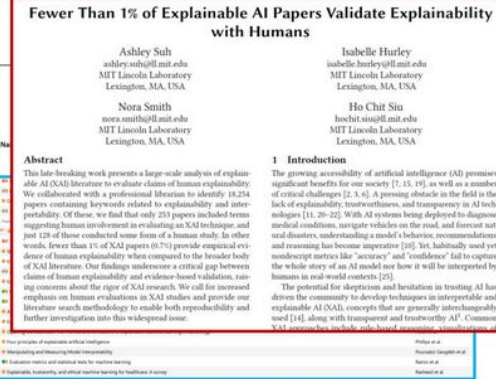
- Lack of consensus and standardized methods
- Gap between theory and human evaluation
- Explainable AI usability study



White-box also called “glass-box”, “model-specific”, or “interpretable models” as the requirement for an interpretable model). White-box models often follow a limited set of rules or parameters, enabling humans to understand their underlying functionality.



Black-box (also called “model-agnostic”) models. black-box models use hundreds or even thousands of decision trees, e.g. “random forests”, or billions of parameters in deep learning models, making it challenging for humans to understand and explain their decisions.



Model Cards for Model Reporting

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Model Cards

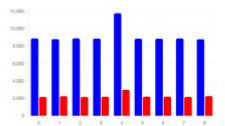
Model Card - EEG Emotion Classifier (v 0.1.0)

Overview

This model classifies EEG signals into **negative**, **neutral**, and **positive** classes. It predicts emotions such as *anger*, *disgust*, *fear*, *amusement*, *joy*, and more. Additionally, it provides uncertainty estimation of predictions with confidence intervals, hyperparameters, and performance metrics.

Dataset

- **Dataset:** FACED Dataset
- **Number of Classes:** 3
- **Ground Truth:** Negative (0), Neutral (1), Positive (2)
- **Training/Validation Split:** 80:20
- **Preprocessing:** Band-Pass Filtering, Common Spatial Patterns (CSP), Normalization

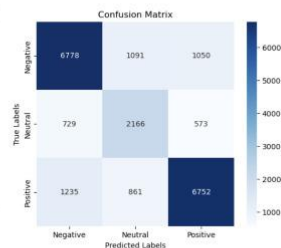


Model Details

- **Input:** 30-channel EEG time-series data
- **Output:** Class labels, Confidence intervals
- **Model Type:** CNN (Xception-based)
- **Learning Rate:** 0.001
- **Batch Size:** 32
- **Parameters:** 0.56M trainable parameters

Performance

The model's performance is evaluated on an independent test set. Below is a summary of the performance by confusion matrix.

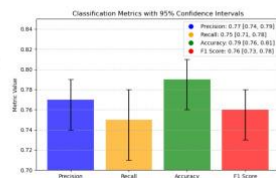


Limitations

- The dataset exhibits class imbalance (male-to-female ratio 3:1), which might introduce bias.
- Noisy EEG channels and missing data can degrade classification performance.

Uncertainty

Confidence intervals provide insights into model reliability. Below, the performance metrics are reported with 95% confidence intervals.



ABSTRACT

Trained machine learning models are increasingly used to perform high-impact tasks in areas such as law enforcement, medicine, education, and employment. In order to clarify the intended use cases of machine learning models and minimize their usage in contexts for which they are not well suited, we recommend that released models be accompanied by documentation detailing their performance characteristics. In this paper, we propose a framework that we call model cards, to encourage such transparent model reporting.

KEYWORDS

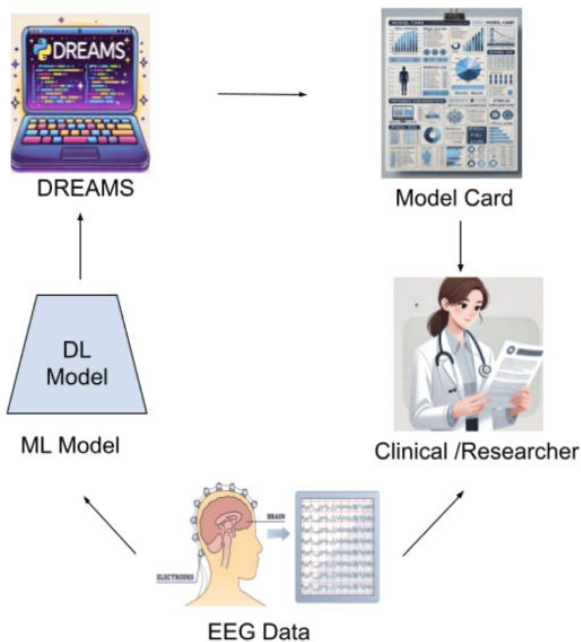
datasheets, model cards, documentation, disaggregated evaluation, fairness evaluation, ML model evaluation, ethical considerations

ACM Reference Format:

Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, Timnit Gebru. 2019. Model Cards for Model Reporting. In *FAT* '19: Conference on Fairness, Accountability, and Transparency*, January 29–31, 2019, Atlanta, GA, USA. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3287560>.

- Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD)
- Google
- HuggingFace

Why DREAMS?



We hypothesize that:

- A domain-specific model card frameworks (DREAMS) tailored for EEG-based deep learning can improve the reporting of performance metrics, dataset biases, and model uncertainty compared to generic model documentation approaches.
- Automated documentation of EEG preprocessing steps, dataset details, and evaluation results will reduce inconsistencies and improve reproducibility in deep learning-based EEG research.
- A standardized model card structure incorporating EEG-specific metadata can facilitate better interpretability and regulatory compliance in healthcare AI applications.

Fig. 1. Overview of the potential role and usefulness of DREAMS for EEG data analysis in the context of clinical research.

Table
Comp
Feat
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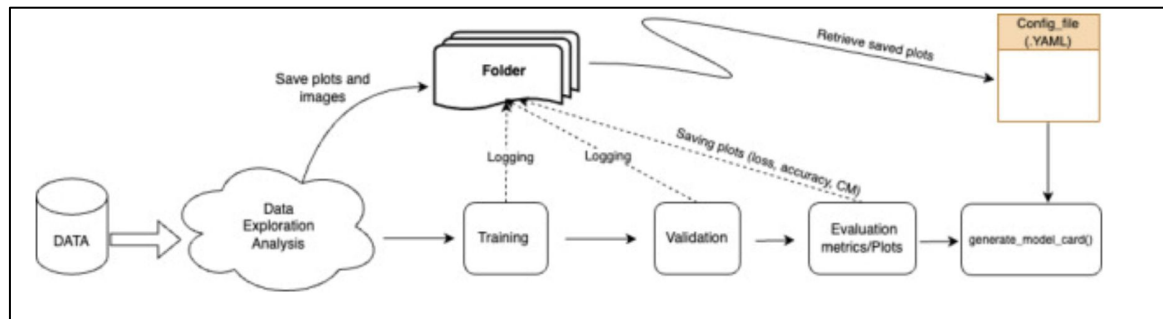
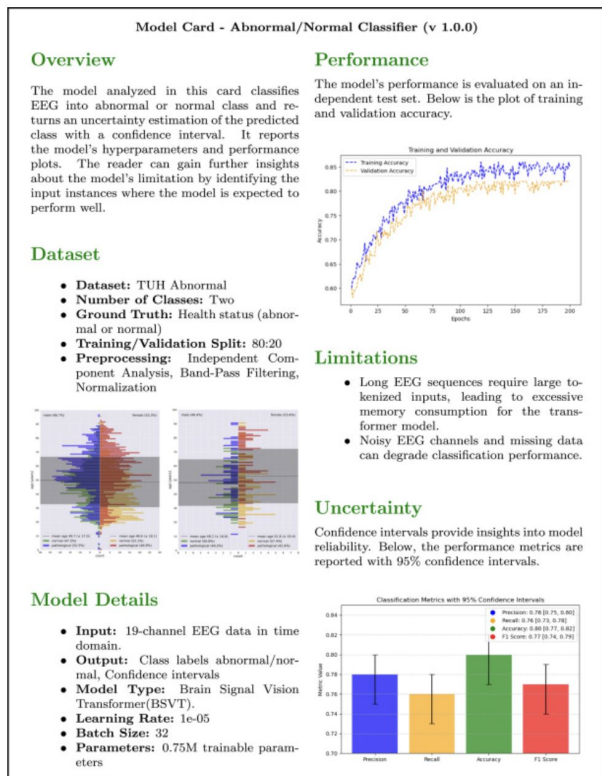
Uniqueness

- From the snippet of the paper on the right, it can be seen that certain datasets for healthcare, including EEG data, are **inherently isolated and skewed because of the nature of how they are collected.**
- We can use DREAMS to actively explore these biases within datasets in healthcare and see what solution can be proposed to reduce these biases

Datasets Used in MLH papers

We examined the breakdown of the kinds of datasets that are used and released in MLH papers more thoroughly (Figure S1). Public, commonly used datasets are rare in the health domain. The vast majority of MLH papers utilized private datasets, which were only available to the authors of the paper itself or a small group containing the authors (and thus tended to be only used by at most one paper in our analysis), or self-collected/simulated datasets designed by the authors of the MLH papers. There is also skewing in the common publicly used datasets, with Medical Information Mart for Intensive Care III (MIMIC-III, <https://mimic.physionet.org/>) taking the lead with 26 recorded usages, followed by the Alzheimer's Disease Neuroimaging Initiative (ADNI, <http://adni.loni.usc.edu/>) with 8, and the rest used at most 3 times. This overuse of MIMIC-III suggests a troubling reliance on a small set of isolated, relatively static resources.

DREAMS documentation



- Dreams aims to provide a comprehensive outlook on biases within training models based on its datasets.
- We are going to take this idea and apply to various datasets and training models and see how different this is compared to previous models

Methodology

1. Data selection and processing

- a. Current Datasets:
 - i. Motor Movement and Imagery:
<https://physionet.org/content/eegmmidb/1.0.0/S001/#files-panel>
 - ii. Eye Movement: https://www.bbc.de/competition/iv/desc_4.pdf
- b. OPS (online pre-alignment strategy) data cleaning?

2. Models and Architectures: → **we are here**

- a. EEGNet - performance on a benchmark model
- b. DeepConvNet - performance on larger more complex models
- c. ShallowConvNet - performance of lightweight networks

3. Train Models on the Datasets

4. Generate Model Cards through DREAMS

- a. Accuracy, Precision and Recall,
- b. Confusion Matrix

5. Compare Results against other Model Cards

- a. Google Model Cards

6. Usability Interviews?

Usability Framework

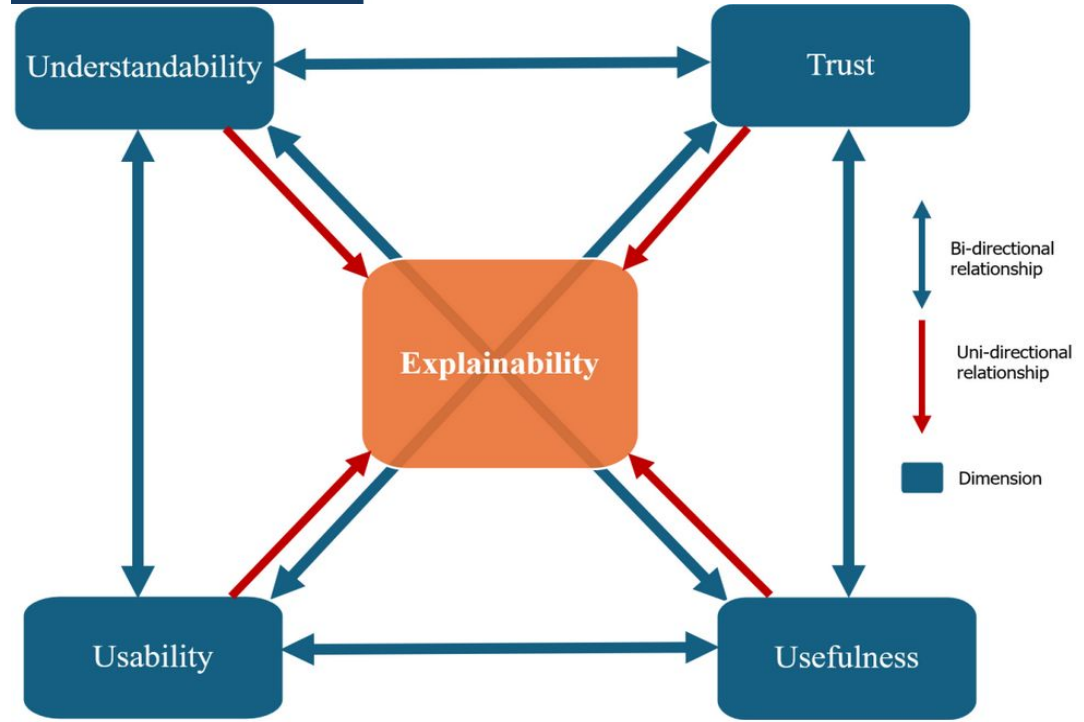
We aim to research the explainability by taking a user-centric design approach.

- We will test and deploy a tool built on a novel ML explainability framework.
- We will conduct a usability study of the tool with clinicians, utilizing the System Explainability Scale (SES) metric.



Human Evaluation of Explainability of an AI-driven Clinical Decision Support System

Mohammad Golam Kibria, PhD, MPH, MBA



Understandability and Usability are Key Themes (Dimensions) in User Testing Sessions

Binomial sign test with Bonferroni-adjusted p-values to account for multiple comparisons

Theme (Dimension)	Successes	Trials	p-value	Adjusted p-value	CI Lower	CI Upper
Understandability	12	13	0.003	0.012	0.640	0.998
Trust	3	12	0.146	0.584	0.055	0.572
Usability	11	13	0.022	0.088	0.546	0.981
Usefulness	6	9	0.508	2.032	0.299	0.925

- Understandability and usability emerged as statistically significant themes in the user testing sessions.
- Trust was less prevalent in the user testing phase when they actually used the AI tool, the clinicians mentioned it less.
- Usefulness remained relatively low in both phases, indicating that participants are less concerned about this area.

Possible Additional Datasets

2.6. Experiments

Within-subject, cross-subject, and cross-dataset experiments were carried out in this work. In the within-subject experiment, we compared the subject-specific performance of both traditional methods and deep learning models. In the cross-subject experiment, the unsupervised transfer ability of two deep learning models was verified in a single dataset. In the cross-dataset experiment, we further validated deep learning models on different datasets with/without online re-centering transformation. The Wilcoxon signed rank test was used to compare the performance of different methods.

In the within-subject experiment, a Linear Discriminant Analysis classifier was used for CSP and FBCSP feature extraction methods. In CSP method, the number of selected spatial filters was set to two. The filter bank of FBCSP is 4–9, 8–15, and 15–30 Hz. The number of selected spatial filters in FBCSP was set to

column method ($p < 0.05$). Under without PS condition, FBCSP and ShallowNet were significantly better than CSP and MDRM. Under with PS condition, all methods were significantly better than CSP. FBCSP, ShallowNet and EEGNet were significantly better than MDRM, whereas no significant differences were observed between deep learning models and FBCSP.

3.2. Cross-Subject Classification Results

Figures 3, 4 show results of cross-subject classification on eight datasets for ShallowNet and EEGNet, respectively. The performance of deep learning models without OPS in cross-subject classification was significantly higher than the random level (ShallowNet: $p = 0.008$; EEGNet: $p = 0.008$). ShallowNet with OPS was significantly better than that without OPS (ShallowNet: $p = 0.046$; EEGNet: $p = 0.062$). Specifically, for CNBCIC2019004 dataset, ShallowNet with OPS increased the accuracy by 19.8% and EEGNet with OPS increased the

2. MATERIALS AND METHODS

2.1. Datasets

Eight MI datasets were used in our experiments (Schalk et al., 2004; Leeb et al., 2007; Tangermann et al., 2012; Yi et al., 2014; Zhou et al., 2016; Cho et al., 2017). All datasets are publicly available and details of them are listed in Table 1. CNBCIC2019004 and CBCIC2019004 datasets were downloaded from the 3rd China Brain-computer Interface Competition website. The rest of datasets were downloaded using the MOABB package (Jayaram and Barachant, 2018).

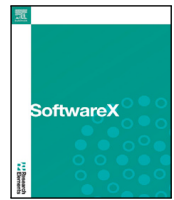
Three channels (C3, CZ, C4) were used in this work. These channels are located on sensorimotor area and exist in all datasets. Only the left-hand and right-hand MI classes were included in our experiments. Each trial was 3 s in length and downsampled to 100 Hz such that the size of a trial was 3×300 . All trials were filtered with a 4-order Butterworth bandpass filter of 3–40 Hz. Zero-phase forward and reverse filtering was implemented using `filter_data()` function in MNE (Gramfort et al., 2013).

For evaluating performance of models, trials were randomly split into training, validation, and test sets. The training set was 80% of the available data. The remaining 20% data were equally partitioned and referred to as validation and test sets. This splitting process was repeated 10 times on each subject, producing 10 different folds.

TABLE 5 | Within-subject Classification accuracies averaged on 10-folds.

	CSP		FBCSP		MDRM		ShallowNet		EEGNet***	
	w/o PS	w/ PS	w/o PS	w/ PS	w/o PS	w/ PS	w/o PS	w/ PS	w/o PS	w/ PS
BNIC2014001	0.68	0.66	0.70	0.72	0.68	0.68	0.76	0.77	0.78	0.79
BNIC2014004	0.70	0.69	0.74	0.75	0.69	0.69	0.79	0.79	0.79	0.80
PhysionetMI	0.56	0.56	0.59	0.61	0.57	0.57	0.53	0.56	0.51	0.56
Cho2017	0.57	0.57	0.60	0.59	0.58	0.58	0.68	0.68	0.65	0.66
Walbo2014	0.66	0.65	0.68	0.69	0.68	0.65	0.75	0.76	0.71	0.74
Zhou2016	0.81	0.82	0.89	0.88	0.80	0.82	0.83	0.87	0.84	0.88
CBCIC2019001	0.57	0.55	0.60	0.60	0.59	0.57	0.66	0.66	0.71	0.71
CBCIC2019004	0.69	0.69	0.74	0.73	0.70	0.70	0.65	0.65	0.62	0.65
Mean	0.65	0.65	0.69	0.70	0.66	0.66	0.71	0.72	0.70	0.72

Stars correspond to *** $p < 0.01$. The boldface shows that the accuracy of method with pre-alignment strategy (w/ PS) is higher than that without pre-alignment strategy (w/o PS).



Original software publication

DREAMS: A python framework for training deep learning models on EEG data with model card reporting for medical applications

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ABSTRACT

Electroencephalography (EEG) provides a non-invasive way to observe brain activity in real time. Deep learning has enhanced EEG analysis, enabling meaningful pattern detection for clinical and research purposes. However, most existing frameworks for EEG data analysis are either focused on preprocessing techniques or deep learning model development, often overlooking the crucial need for structured documentation and model interpretability. In this paper, we introduce DREAMS (Deep REport for AI Models), a Python-based framework designed to generate automated model cards for deep learning models applied to EEG data. Unlike generic model reporting tools, DREAMS is specifically tailored for EEG-based deep learning applications, incorporating domain-specific metadata, preprocessing details, performance metrics, and uncertainty quantification. The framework seamlessly integrates with deep learning pipelines, providing structured YAML-based documentation. We evaluate DREAMS through two case studies: an EEG emotion classification task using the FACED dataset and an abnormal EEG classification task using the Temple University Hospital (TUH) Abnormal dataset. These evaluations demonstrate how the generated model card enhances transparency by documenting model performance, dataset biases, and interpretability limitations. Unlike existing model documentation approaches, DREAMS provides visualized performance metrics, dataset alignment details, and model uncertainty estimations, making it a valuable tool for researchers and clinicians working with EEG-based AI. The source code for DREAMS is open-source, facilitating broad adoption in healthcare AI, research, and ethical AI development.

Code metadata

Current code version
Permanent link to code/repository used for this code version
Legal Code License
Code versioning system used
Software code languages, tools, and services used
If available Link to developer documentation/manual
Support email for questions

v0.0.3
<https://github.com/ElsevierSoftwareX/SOFTX-D-24-00509>
Released under BSD 3-Clause License
git
Python, Java, CSS, SCSS
<https://dreams-mc.readthedocs.io/en/latest/index.html>
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1. Introduction

Electroencephalography (EEG)-based deep learning has demonstrated significant potential in various applications, including brain-computer interfaces (BCI), emotion recognition, and neurological disorder diagnosis [1–3]. Despite these advancements, a key challenge in

the development and deployment of deep learning models for EEG applications is the lack of structured and standardized model documentation. Models designed for EEG based applications are highly sensitive to variations in dataset preprocessing, feature extraction techniques,

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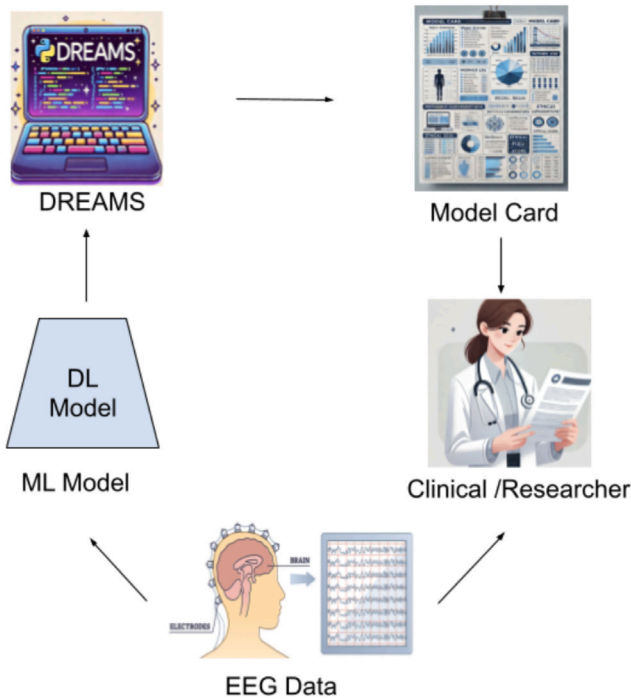


Fig. 1. Overview of the potential role and usefulness of DREAMS for EEG data analysis in the context of clinical research.

and model architectures, making comparability and reproducibility challenging [4].

Reproducibility is a fundamental requirement in healthcare applications of machine learning (ML) models, yet EEG-based deep learning research and applications continues to suffer from inconsistent and non-transparent model reporting practices. Many studies report only high-level performance metrics (e.g., accuracy, F1-score) while omitting crucial details such as preprocessing pipelines, hyperparameter configurations, and prediction uncertainties. The absence of structured documentation not only hinders reproducibility but also complicates model benchmarking and comparative analysis, limiting the ability of researchers to validate findings across different datasets and experimental settings. This challenge becomes even more critical in clinical applications, where a lack of reproducibility can impede regulatory approval, deployment, and real-world patient outcomes. Addressing these gaps requires a standardized and automated model documentation framework that ensures transparent, structured, and reproducible reporting of EEG-based deep learning models.

One of the key challenges in deploying deep learning models is their inherent uncertainty in predictions, often leading to overconfidence in incorrect classifications [5,6]. This issue is particularly critical in EEG-based deep learning, where high signal variability, inter-subject differences, and noise hinder model generalization capacity [7]. In clinical settings, failing to quantify uncertainty can result in misdiagnoses or incorrect treatments, posing high risks [4].

Uncertainty estimation enhances risk-aware decision-making, allowing models to flag low-confidence predictions for further review instead of making potentially harmful automatic decisions. For example, an EEG-based seizure detection system with uncertainty awareness can alert clinicians when a prediction is inconclusive, prompting manual verification. Integrating uncertainty estimation into structured model documentation, such as model cards, ensures greater transparency, interpretability, and trustworthiness in EEG-based healthcare AI [8].

To address these challenges, we introduce DREAMS, a Python-based model card framework specifically designed for EEG-based deep learning models. DREAMS provides automated documentation covering

key aspects such as dataset characteristics, preprocessing techniques, model architecture, performance evaluation with confidence intervals, and version tracking.

Our main contributions are:

1. We propose DREAMS, a structured model card framework for EEG deep learning, enabling comprehensive documentation of model performance, preprocessing pipelines, and uncertainty estimation.
2. We introduce confidence interval-based uncertainty reporting, improving model interpretability and helping researchers assess the prediction reliability of the models.
3. We automate model documentation with version tracking, allowing for efficient benchmarking and reproducibility across different model versions and EEG datasets.
4. We demonstrate the framework's applicability in EEG-based classification tasks, showcasing its potential for broader EEG-based ML applications.

Fig. 1 sketches DREAMS role in the context of clinical research based in EEG data. The [code metadata](#) table provides the main metadata of the code of DREAMS.

2. Related work

Some previous works must be highlighted, which will better contextualize and motivate the novelty of DREAMS. Moons et al. [9] introduced the Transparent Reporting of Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD) statement, which aimed to enhance the reporting of prediction models. The TRIPOD statement is a checklist comprising items essential for the transparent reporting of prediction model studies. This explanation document details the rationale behind each item, clarifies their meanings, and serves as a valuable reference for considerations when reporting the design and analysis of prediction model studies.

Sendak et al. [10] introduced the Model Facts label, designed for clinicians who make decisions supported by machine learning models. Its purpose is to collect relevant, actionable information into a single page. Its major sections include the model name, locale and version, a summary of the model, the mechanism of risk score calculation, validation and performance metrics, usage directions, warnings and additional information.

Crisan et al. [11] explored interactive model cards, which enhance traditional static model cards by allowing users to explore documentation and interact with models. It includes a conceptual study with ML, NLP and AI Ethics experts, followed by an evaluative study with non-expert analysts.

Mitchell et al. [8] presented a framework for transparent model reporting, primarily targeting human-centered models in computer vision and natural language processing, though it can be applied to any machine learning model. They illustrated the concept using model cards for two examples: one that detects smiling faces in images and another that identifies toxic comments in text. All of the architectures mentioned have certain limitations: they may oversimplify complex details and are often tailored to specific settings. Some of these approaches require technical infrastructure for implementation and others are presented as guidelines on particular domains, limiting their broader applicability.

2.1. Comparison with existing model documentation frameworks

The increasing adoption of deep learning in EEG analysis necessitates structured documentation tools that ensure transparency, reproducibility, and ethical compliance. While existing model documentation frameworks offer general-purpose solutions, they often lack domain-specific adaptability for EEG-based AI. This section compares

DREAMS with prominent model documentation approaches, highlighting its unique contributions.

Several model documentation framework exists, but they are primarily designed for general machine learning models rather than the specialized needs of EEG-based deep learning. One such framework is the Google's Model Card Toolkit (MCT) [12] that provides structured model reporting, but it is heavily integrated with TensorFlow Extended (TFX) and lacks direct support for EEG preprocessing metadata or domain-specific performance metrics. Similarly, Hugging Face Model Cards [13] have become a standard for documenting NLP and computer vision models, but they do not accommodate the interpretability challenges, uncertainty quantification, or ethical considerations unique to EEG-based deep learning. While the traditional manual model documentation methods [9] (e.g., static reports, research papers) is flexible but lacks automation, standardization, and reproducibility, making it time-consuming and inconsistent across studies. Without a structured and automated approach, documenting EEG-based deep learning models remains inefficient and prone to inconsistencies, underscoring the need for a dedicated framework like DREAMS.

Unlike these frameworks, DREAMS is specifically designed for EEG-based deep learning models and offers several key advantages, namely:

- **EEG-Specific Adaptation.** DREAMS includes EEG preprocessing details, dataset alignment reports, and uncertainty estimation, ensuring domain-relevant transparency.
- **Automated Documentation.** Unlike manual reporting, DREAMS automatically logs model parameters, training details, and performance metrics, reducing human error and improving reproducibility.
- **Framework Agnostic.** Unlike Google's Model Card Toolkit, which is tied to TensorFlow, DREAMS works with any deep learning framework (e.g., PyTorch, Keras, TensorFlow).
- **Enhanced Interpretability.** DREAMS provides visualized model performance metrics, confidence intervals, and dataset biases, making it more informative than text-heavy model cards.
- **Ethical and Regulatory Compliance.** By documenting bias risks, dataset composition, and interpretability challenges, DREAMS aligns with best practices in responsible AI for medical applications.

Table 1 summarizes the differences between DREAMS and existing model documentation approaches.

3. Software description

To illustrate the concept of DREAMS, we present a case study in which model cards (MC) are utilized to highlight the key aspects of the data, model, and performance metrics, as well as the limitations encountered in the study. MCs serve as a detailed documentation tool, emphasizing the characteristics and context of the data used, the specific features and architecture of the model and reports its performance. Moreover, it can include information about limitations or challenges identified during the research. The use of MCs when developing DL underscores the importance of transparency and accountability in model development and deployment.

3.1. Software architecture

DREAMS, developed as a Python package, offers a comprehensive framework aimed at simplifying the process of creating comprehensive model cards for deep learning models. Model cards (MC) serve as vital documentation tools, outlining the performance metrics, usage guidelines and limitations of AI models. They play a pivotal role in enhancing transparency and promoting the ethical utilization of AI technologies.

The DREAMS architecture is modular, comprising several key components. It starts with exploratory data analysis (EDA) on the provided

dataset, with the visualizations saved in a designated folder. As the model progresses through training and validation, this folder is further enriched with additional plots, such as loss, accuracy and confusion matrix. Finally, the model card function is invoked, using a configuration file that contains all the necessary details to generate the model card. An overview of the architecture of DREAMS' library is shown in Fig. 2.

3.2. Software functionalities

3.2.1. Data preprocessing

The pipeline typically begins with data collection and preprocessing, where the raw data is gathered from various sources and prepared for model input. Preprocessing steps may include normalization, data augmentation and transformation into formats that are compatible with the model, such as converting images to tensors or encoding text data. Then, the data is usually divided into training and validation sets. The training set is used to train the model, while the validation set assists fine-tune the hyperparameters of the model. To ensure that every step in this process is well-documented and reproducible, we systematically store all critical details about the data, including information on preprocessing, in a designated folder. In a YAML file, we define the exact path to this folder, ensuring that the information is easily accessible and organized for future reference or use. This DREAMS framework setup not only facilitates transparency but also enables efficient tracking of data-related processes throughout the entire model development pipeline.

3.2.2. Model training and evaluation

The training pipeline of a deep learning model is a comprehensive process that transforms raw data into a well-trained model capable of making accurate predictions or classifications. During the training phase, the model processes batches of data iteratively, adjusting its internal parameters to minimize the error between its predictions and the actual outcomes. This is achieved through an optimization algorithm like stochastic gradient descent, which updates the model's parameters based on the gradients of the loss function. Throughout the training pipeline, performance metrics such as accuracy, loss, precision, recall, and F1-score are monitored to assess the model's progress. Visualization tools such as confusion matrices are often used to gain deeper insights into the model's performance. The model is then ready for inference, where it can be used to make predictions on new, unseen data.

At this stage, we meticulously document all the paths to vital information related to both training and evaluation results within the YAML file. This DREAMS comprehensive record encompasses not only paths to graphics but also to other essential data, ensuring that every critical component is easily accessible and well-organized. This approach ensures efficient management and retrieval of information throughout the training and evaluation processes.

3.2.3. Model card generation

Once all the necessary information for generating the model card has been gathered, we proceed to invoke the `generate_modelcard()` function. This function requires three input parameters: the path to the YAML configuration file, which contains the essential details for the model card; the output path, where the generated model card will be saved; and the version number of the model card, which helps track its iterations and updates.

An example of the code snippet using `generate_modelcard()` function is as follows:

```
1 # model_card library
2 from dreams_mc.make_model_card import
   generate_modelcard
3
4 config_file_path = './config.yaml'
5 output_path = './logs/model_card.html'
6 version_num = '1.0'
```

Table 1
Comparison of model documentation frameworks.

Feature	Google MCT	Hugging Face	Manual Docs	DREAMS
Designed for EEG Models?	✗ (No)	✗ (No)	(Limited)	✓ (Yes)
Automated Model Card Generation	(Partial, TFX)	✗ (No)	✗ (No)	✓ (Yes)
Integration with DL Frameworks	TensorFlow Only	PyTorch /TensorFlow	✗ (No)	Works with Any DL Framework
Dataset Transparency	Basic Metadata	Basic Metadata	(Manual)	EEG-Specific Metadata
Performance Metrics Logging	(Limited)	✓ (Yes) (for NLP)	✗ (No)	✓ (Yes, Integrated)
Uncertainty Estimation	✗ (No)	✗ (No)	✗ (No)	✓ (Yes) (Confidence Intervals)
Ethical Considerations	✓ (Encouraged)	✓ (Encouraged)	Researcher-Dependent	✓ (EEG Bias Reporting)
Visualization of Results	✗ (No)	✗ (No)	✗ (No)	✓ (Performance Plots)
Reproducibility	(Limited, TFX Required)	✗ (No Tracking)	✗ (No Tracking)	✓ (YAML-Based Logging)
Customization	(Limited)	(Limited to NLP)	✗ (No Format)	✓ (Flexible YAML)
Open Source?	✓ (Yes)	✓ (Yes)	✗ (No)	✓ (Yes)

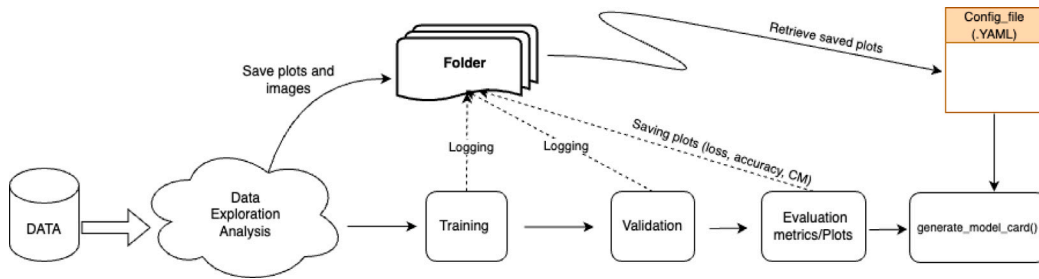


Fig. 2. Workflow of the DREAMS model card generation process: The diagram illustrates the end-to-end pipeline, starting from data collection and exploration, followed by training and validation phases, and progressing to evaluation metrics and plot generation. Key outputs, including plots and logs, are saved in a designated folder, which are then retrieved by the configuration file (.YAML). Finally, the `generate_model_card()` function utilizes the configuration and saved assets to automatically generate the structured model card.

```

7
8 # Generate the model card
9 generate_modelcard(config_file_path,
  output_path, version_num)

```

4. Hypothesis and limitations

The primary hypothesis of this work is that structured documentation of deep learning models through *automated model cards* can significantly enhance *transparency, reproducibility, and ethical compliance* in EEG-based deep learning. Specifically, we hypothesize that:

- A domain-specific model card frameworks (DREAMS) tailored for EEG-based deep learning can improve the reporting of performance metrics, dataset biases, and model uncertainty compared to generic model documentation approaches.
- Automated documentation of EEG preprocessing steps, dataset details, and evaluation results will reduce inconsistencies and improve reproducibility in deep learning-based EEG research.
- A standardized model card structure incorporating EEG-specific metadata can facilitate better interpretability and regulatory compliance in healthcare AI applications.

To support this hypothesis, we make some assumptions regarding the use of EEG data and deep learning models:

- The quality of model documentation is a critical factor affecting the interpretability and trustworthiness of EEG-based deep learning models.
- The datasets used in this study (FACED and CAUEEG) are representative of real-world EEG classification tasks, allowing generalization of results to similar applications.
- The reported performance metrics (e.g., accuracy, F1-score, uncertainty estimates) provide meaningful insights into model effectiveness and limitations, which are accurately captured in the generated model cards.
- Researchers and clinicians will benefit from structured model documentation without significant modifications to existing deep learning workflows.

4.1. Limitations

Despite the advantages of DREAMS, we acknowledge some limitations. The framework's effectiveness depends on the EEG *data quality* and preprocessing techniques. Poorly preprocessed or noisy EEG signals may introduce biases that cannot be fully mitigated through documentation alone. Additionally, while DREAMS automates model documentation, the logging and visualization processes introduce *computational overhead*, which may not be ideal for real-time EEG applications requiring low-latency performance. Another limitation is the framework's current level of *clinical validation*; while it is designed for research

and development, its direct applicability in clinical environments requires further evaluation with real-world patient data and potential regulatory compliance assessments. Furthermore, although DREAMS is framework-agnostic, its implementation is *primarily optimized* for Python-based deep learning libraries such as PyTorch and TensorFlow, limiting its immediate usability for researchers working with other AI ecosystems. Lastly, while DREAMS promotes transparency by documenting model biases and performance limitations, it does not eliminate the *ethical concerns* associated with EEG-based deep learning models. Ensuring fair data collection, privacy protection, and algorithmic fairness remains the responsibility of researchers and developers implementing the framework. By recognizing these challenges, we highlight areas for future improvements and refinements, ensuring DREAMS continues to evolve as a robust and practical model documentation tool for EEG-based AI applications.

5. Illustrative examples on EEG data

We provide examples of model cards for two models: an EEG-based emotion classifier and an EEG dementia classification model.

5.1. Emotion classifier

We present a model card for an EEG-based emotion classifier trained on the Finer-grained Affective Computing EEG Dataset (FACED) [14], which is compatible with TorchEEG.¹ The dataset consists of EEG recordings from 123 subjects using 32 electrodes positioned according to the international 10–20 system. During the recording sessions, participants viewed 28 different video clips designed to elicit various emotional responses. These video clips spanned nine distinct emotion categories: amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, and neutral emotion. We applied band-pass filtering, common spatial pattern filtering, and normalization to the EEG data.

We applied a deep learning model called TSception, a multi-scale convolutional neural network designed to classify emotions from EEG data. It incorporates dynamic temporal layers, asymmetric spatial layers and high-level fusion layers, enabling the model to learn discriminative representations across both time and channel dimensions simultaneously. The model described in this example classifies EEG data into three main categories: negative (0), which includes emotions such as anger, disgust, fear, and sadness; neutral (1); and positive (2), which includes emotions like amusement, inspiration, joy, and tenderness.

The model card provides a structured evaluation of model performance, including confusion matrix analysis, from which true positive (TP), false positive (FP), true negative (TN), and false negative (FN) rates are derived to assess classification performance. Additionally, it includes uncertainty estimation, offering insights into the confidence levels of model predictions. Furthermore, the model card provides comprehensive documentation, outlining key aspects of the dataset characteristics, preprocessing steps applied to EEG signals, and an overview of the deep learning model for emotion classification.

Fig. 3 illustrates the prototype of the model card, highlighting its structured approach to model transparency and interpretability. Notably, the model card is fully customizable and can be extended to incorporate additional domain-specific details, depending on the specific application requirements.

5.2. Abnormal classifier

We provide a second illustrative model card using the publicly available abnormal EEG dataset from the Temple University Hospital (TUH) v2.0.0 [15]. The dataset includes 2993 recordings from 2329 different subjects. The recordings are labeled as normal (1385 recordings) or abnormal (998 recordings). The training set contains 2717 recordings and the evaluation set holds 276 recordings. We applied band pass filtering, Independent Component Analysis (ICA), Normalization as the preprocessing steps on the input dataset.

A transformer-based model, BSVT [16], was employed for the classification task, with its performance and prediction uncertainty systematically documented in the model card. Fig. 4 presents an example of the model card, showcasing how variations in model architecture and performance evaluation can be structured within a standardized format. Notably, if hyperparameters are modified or an alternative model is selected, the model card can accommodate these updates by specifying the version number and associated changes. Furthermore, the model card is highly customizable, allowing for the inclusion of additional sections as needed. For instance, a recommendation section could be added to provide additional information on model deployment, usage considerations, or the necessity for further validation and testing.

5.3. Pipeline for model card generation

This section details the step-by-step process for training a deep learning model using pytorch framework, evaluating its performance, and ultimately generating a structured model card. The pipeline includes data preprocessing, model training, validation, performance analysis, and automated documentation using a model card.

Data Preprocessing and Preparation The EEG data undergoes online transformation into tensor format and a 2D representation for compatibility with deep learning models. The labels are adjusted accordingly, and the dataset is split into training and validation sets. To efficiently manage data batching and shuffling, we create DataLoaders for both training and validation. The implementation is as follows:

```
1 data_folder= "./processed_data_Face/
   Processed_data"
2
3 dataset = Dataset(root_path= data_folder,
4                   online_transform=
5                   transforms.Compose(
6                       [transforms.ToTensor
7                       (),
8                       transforms.To2d()] ),
9                   label_transform=
10                  transforms.Compose([
11                      transforms.Select('
12                      valence'),
13                      transforms.Lambda(
14                          lambda x: x + 1)
15                      ]))
16
17 # Dataset Split
18 train_dataset, val_dataset = random_split(
19     dataset, [num_train_samples,
20             num_val_samples])
21
22 # DataLoaders creation
23 train_loader = DataLoader(train_dataset,
24                           batch_size=batch_size, shuffle=True)
25 val_loader = DataLoader(val_dataset,
26                         batch_size=batch_size, shuffle=False)
```

Model Setup and Configuration At this stage of the pipeline, the model is initialized and configured with the necessary parameters. The model is instantiated using the *Model function*, specifying

¹ <https://torcheeg.readthedocs.io/en/latest/>



Fig. 3. An illustrative example of model card for EEG emotion classification. This model card summarizes the dataset, model details, performance metrics, uncertainty of predictions and limitations for an EEG-based emotion classification system.

key hyperparameters for instance, in our case the number of classes, electrodes, sampling rate, and architectural components. The following code snippet demonstrates the initialization:

```

1
2 model = Model(num_classes=9,num_electrodes
               =30,
3               sampling_rate=250,num_T=15,
4               num_S=15, hid_channels=32,
               dropout=0.5)

```

Training and Validation Process We proceed to configure the training and validation processes utilizing PyTorch. This involves setting up the necessary parameters and components for training the model, such as defining the loss functions, optimizers, and evaluation metrics. We establish the procedures for validating the model's performance, including managing the data flow, monitoring metrics, and adjusting the training process as needed to ensure effective learning and evaluation. A standard implement for the training and validation configuration is:

```

1 # Model Training
2 valid_acc_max = 0.0

```

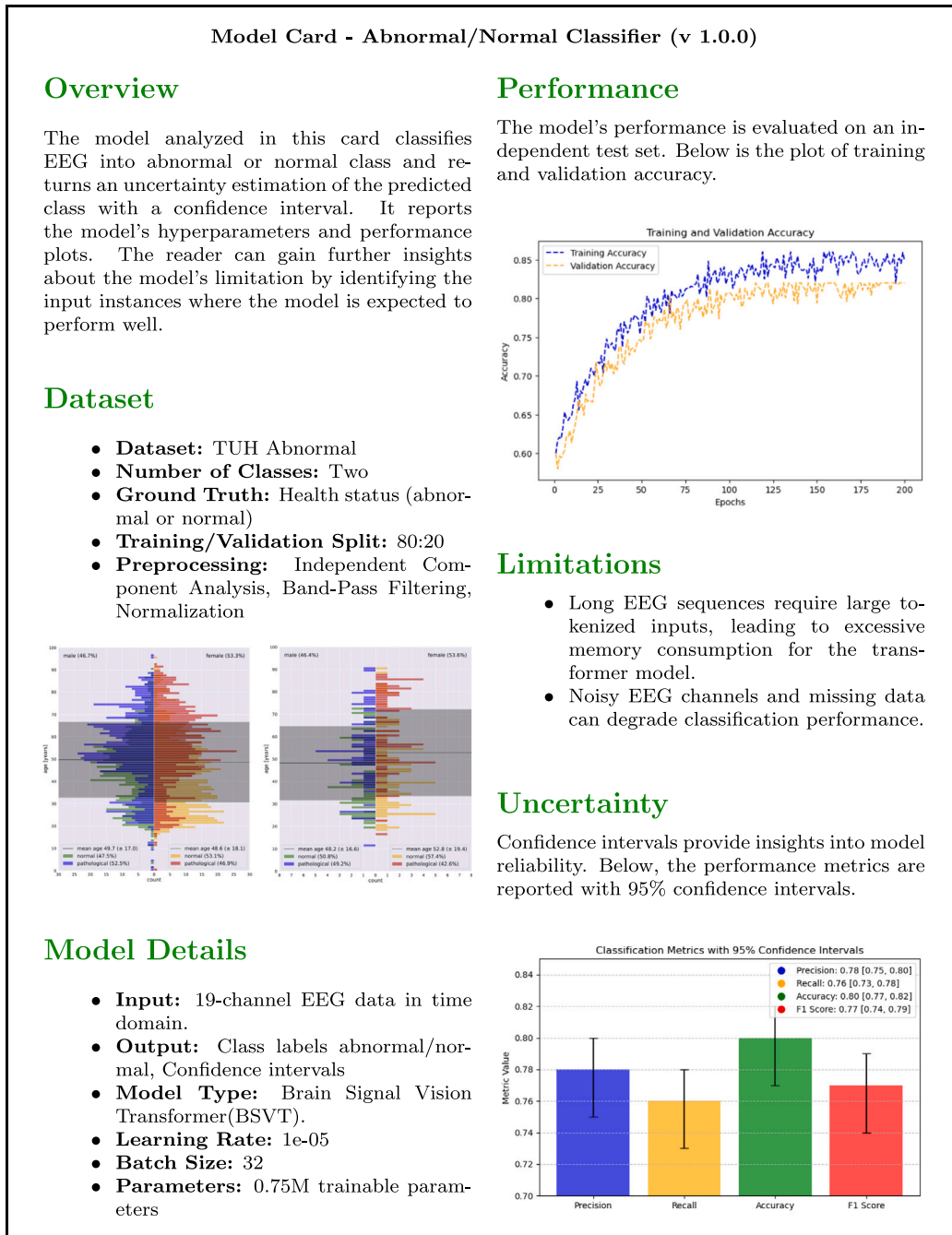


Fig. 4. An illustrative example of model card for EEG abnormal/normal classification. This model card summarizes the dataset, model details, performance metrics, uncertainty of predictions and limitations for the classification of abnormal and normal EEG signals.

```

3 trained_model, accuracy_stats, loss_stats =
  train(220, valid_acc_max, model,
    optimizer, criterion, scheduler,
    train_loader, val_loader, "./logs/
    current_checkpoint.pt", "./logs/
    best_model.pt", start_epoch=1)

4
5 # plotting and saving training and
  validation plots
6 plot_training_validation_stats(
  accuracy_stats, loss_stats, save_dir='./
  logs')

7
8 # Load the best model for evaluation
9 checkpoint_path = './logs/best_model.pt'

10 checkpoint = torch.load(checkpoint_path,
  map_location=torch.device('cpu'))
11 model.load_state_dict(checkpoint['state_dict'
  '])

Performance Evaluation and Metrics Visualization To assess the
model's performance, we generate a confusion matrix and compute
key evaluation metrics, such as accuracy, precision, recall, and F1-
score. Additionally, uncertainty estimation is applied to quantify the
confidence of predictions.

1 plot_metrics_table(results, model_name='
  TSception', save_path="./logs/table.png")

```

```

2 plot_confusion_matrix(results['
    confusion_matrix'], class_names,
    save_path="./logs/cm.png")
3 plot_confidence_intervals(precision=results[
    'precision'], recall=results['recall']
    ...

```

Model card Generation Lastly, we call the `generate_model_card()` function, providing it with the path to the YAML configuration file,² the output path for saving the model card, and the version number of the model card as shown below:

```

1 # Generating model card
2 print("Generating Model Card ...")
3 config_file_path = './config.yaml'
4 output_path = './logs/model_card.html'
5 version_num = '1.0'
6 generate_modelcard(config_file_path,
    output_path, version_num)

```

Figs. 3 and 4 provide the illustrative example of the generated model cards following the above describe pipeline. This model card encapsulates the key information and metrics resulting from the model's training and evaluation processes, offering a comprehensive summary of the model's performance, data characteristics, and other relevant details.

6. Impacts

In the context of EEG applications, DREAMS offers a smooth solution for generating comprehensive model cards, providing detailed sections that include overviews of EEG analysis projects, dataset descriptions specific to EEG recordings, results with corresponding plots (such as EEG waveforms or brain activity patterns), uncertainty estimations related to signal interpretation, and relevant references. Additionally, users can customize the content of these model cards using a YAML configuration file, enabling adaptability to various EEG-based projects and models, ensuring flexibility in presenting key insights and findings.

DREAMS is entirely model-agnostic, deliberately engineered to be compatible with any type of model. Its design ensures versatility, allowing it to generate model cards regardless of the underlying model architecture or type. This flexibility is a fundamental aspect of DREAMS, enabling it to adapt to various machine learning and deep learning models, including but not limited to convolutional neural networks (CNNs), recurrent neural networks (RNNs), transformer models and random forests. Whether it is a classification, regression or any other type of model, DREAMS generates comprehensive model cards, providing valuable insights and documentation irrespective of the model's complexity or structure. The model-agnostic nature of DREAMS enables users to apply it across a diverse array of projects and applications, ensuring its relevance and usefulness in various machine-learning endeavors.

This versatility allows DREAMS to find applicability across a wide range of domains where transparency and ethical use of AI are paramount. For instance, within the healthcare sector [17,18], DREAMS could be utilized to document the performance and limitations of deep learning models used for medical analysis. Through the generation of detailed model cards, healthcare practitioners and regulatory bodies can gain insights into the model's accuracy, potential biases, and areas of uncertainty, facilitating informed decision-making and ensuring patient safety.

Similarly, in the financial sector [19,20], DREAMS can aid in documenting AI models used for credit risk assessment or fraud detection, providing transparency into the model's decision-making process and

potential biases. This transparency is essential for ensuring fairness and accountability in algorithmic decision-making processes.

Furthermore, in intelligent transportation systems [21] such as, autonomous vehicles, DREAMS can be employed to document the performance and safety considerations of deep learning models used for object detection and navigation, promoting trust and transparency among regulators and the general public.

DREAMS can be effortlessly installed and utilized via PyPI, ensuring convenient access for users. The project's code has been shared on GitHub, allowing for collaborative contributions and enhancements from the community. As an open-source initiative, users can extend and customize the software to meet their specific requirements. Comprehensive documentation is available in English to assist users in understanding and utilizing the software effectively. Note that, the maintenance and upkeep of the software are managed by one of the authors, ensuring ongoing support and reliability.

In essence, DREAMS enables the creation of model cards that hold immense value for AI practitioners. These cards provide vital documentation that enhances transparency, facilitates user understanding, and supports responsible deployment by detailing the operational parameters of the model.

7. Discussions and future work

This work introduces DREAMS, a Python-based model card framework for EEG-based deep learning models, designed to enhance transparency, interpretability, and reproducibility. A key advantage of this framework is the integration of automated model documentation, reducing manual effort while ensuring consistency across model iterations. The incorporation of confidence intervals in model evaluation improves interpretability, enabling researchers to quantify prediction reliability and assess model uncertainty. Furthermore, dataset characteristics, model's hyperparameters, preprocessing details, and performance analysis can be systematically documented, ensuring comprehensive and standardized reporting across studies.

Beyond improving reproducibility, DREAMS also addresses critical ethical concerns in EEG-based AI by promoting responsible model documentation. Ethical challenges in deep learning, particularly in EEG-based healthcare and neuroscience applications, include bias in training data, lack of transparency in decision-making, and potential risks in clinical deployment. By explicitly documenting dataset demographics, potential biases, and model limitations, the model card enables researchers to assess fairness and mitigate algorithmic bias. Additionally, DREAMS allows for the inclusion of ethical guidelines, ensuring that researchers disclose data handling practices, consent considerations, and limitations of model generalizability, fostering greater accountability in AI-driven EEG applications.

The customizability of DREAMS ensures that model cards can be extended based on specific use cases, incorporating additional sections such as deployment considerations, ethical guidelines, and regulatory compliance requirements. This flexibility allows researchers to tailor documentation for clinical validation, healthcare deployment, or domain-specific performance evaluation, supporting ethical AI development in high-stakes applications.

While model documentation is a significant step toward improving reproducibility, broader challenges remain in EEG-based deep learning. The variability of EEG signals across individuals introduces inherent uncertainty, making generalization difficult. This necessitates future work on subject-adaptive models and personalized EEG feature extraction. Additionally, the lack of standardized reporting practices in EEG-based deep learning further underscores the need for structured model cards as part of a transparent and ethical AI workflow. By systematically addressing uncertainty, bias, and documentation gaps, DREAMS provides a foundational framework for responsible EEG-based AI research and deployment.

² <https://github.com/LucidJun/DREAMS/tree/main/template>

Future research should explore the integration of explainability techniques, such as saliency maps and attention mechanisms, to provide deeper insights into EEG feature learning. Furthermore, ensuring fairness and bias detection in EEG models, particularly across different demographics, remains an important direction for future study.

In conclusion, this work provides a structured and automated model card framework designed to improve reproducibility, interpretability, and uncertainty-aware evaluation in EEG-based deep learning. We show the integration of DREAMS into deep learning workflow to enhance model transparency by systematically reporting performance metrics such as accuracy, precision, recall, and F1-score with 95% confidence intervals. DREAMS ensures automated documentation with version tracking, reducing manual effort and improving reproducibility across model iterations. By addressing the lack of standardized model documentation, our framework ensures that performance variability, dataset characteristics, and model reliability are systematically reported. It can be extended to various EEG classification tasks, clinical research, and neuroscience applications. Through the introduction of this framework, we aim to establish a standard for transparent and reproducible model reporting, ultimately contributing to responsible AI adoption in EEG research and clinical settings.

CRedit authorship contribution statement

Rabindra Khadka: Writing – review & editing, Writing – original draft, Validation, Software, Conceptualization. **Pedro G. Lind:** Supervision, Methodology, Conceptualization. **Anis Yazidi:** Writing – review & editing, Supervision, Resources. **Asma Belhadi:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Rabindra Khadka reports financial support was provided by European Union. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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